SOLUTIONS SPRINT

Solutions Engineer Intern Assignment — Parts A to D

Wantace Solutions

Document field	Details
Candidate	Golla Narasimha
Submission date	1 October 2026
Local audit subject	SS MS Fabrication Works — publicly listed steel/metal fabrication workshop in the KR Puram / TC Palya area of Bengaluru.
Document status	Completed. Public listing details are real; all internal workflow details are modeled examples, not verified claims.

IMPORTANT SCOPE NOTE

This document uses the publicly listed SS MS Fabrication Works location in KR Puram, Bengaluru, as the proposed field-audit subject. The public address and business category are directory-listed. No owner interview, on-site observation, photographs, internal records or direct owner quotations have been obtained for this document. Therefore, all operational details in Part A are explicitly illustrative—not findings about the named business. Do not represent this sample as a completed field audit. Part B is based on the fictional Vardhan Steelware case supplied in the assignment brief. Part C is the corresponding internal handoff. Part D is a safe, general draft based on the modeled workflow scenario.

PART A — Real Business Audit

Audit status: preliminary desk research complete; field interview and direct observations pending. The assignment requires an owner/manager interview of at least 20 minutes, a hand-drawn process map photo, and at least two permitted photos of tracking methods.

A1. Business profile (publicly listed details only)

Field	Current information / status
Business name	SS MS Fabrication Works (public directory name; confirm exact trading name if conducting a real visit).
Listed address	No. 32/3, Varnasi Road, Jinkethimmanahalli, near Mother Terasa School, TC Palya, KR Puram, Bengaluru, Karnataka 560036, India.
Business activity	Public listings categorize it as a steel/metal fabrication business and mention related fabrication/installation work. These listings do not verify its current service mix.
Public contact	Not included in this report. The assignment permits the owner's phone only in the submission email and only with the owner's permission.
Interviewee / date / duration	No interview conducted for this illustrative sample. A real submission requires a 20-minute or longer owner/manager interview.
Order volume / customer mix	Not publicly verified. The modeled example assumes a small workshop handles custom jobs, but no monthly volume or customer mix is attributed to the named business.
Order channels / tracking tools	Illustrative workflow only: enquiries may arrive by phone/WhatsApp/in person and job progress may be tracked through paper notes or a spreadsheet. Actual channels and tools are unknown.

Public-source notes: business name/address/service categories were found in online directory listings. These sources are useful for identifying a potential audit subject but are not a substitute for an interview or proof of the workshop's internal practices.

A2. Interview scope and actual workflow

Stage to investigate	Questions to answer during the interview	Actual finding
Customer enquiry	Illustrative question: phone, WhatsApp or walk-in; who captures customer and job details?	Illustrative model: enquiry is recorded in a notebook or message thread; not independently verified.
Measurement / specification	Who measures the job and records dimensions, material grade, design and finish?	Illustrative model: measurements are written on a job note and confirmed with the customer before fabrication.

Quotation / approval	How is the quote prepared, approved and recorded? Is an advance required?	Illustrative model: owner prepares a quote, confirms price and delivery expectation, and records any advance.
Order scheduling	How is the job placed in the work queue? Who decides priority and promised date?	Illustrative model: jobs are prioritized verbally or in a simple list; a formal queue is not assumed.
Material / fabrication	How are steel sections/sheets issued, cut, welded and leftovers/scrap tracked?	Illustrative model: material is issued to a job and usable leftovers/scrap are noted inconsistently; this is a discovery hypothesis.
Finishing / quality	Which finishing steps happen in-house or outside? Who checks quality and measurements?	Illustrative model: welding/grinding and any outsourced finish are followed by a visual/dimensional check.
Delivery / installation	Who arranges delivery/installation and confirms completion/payment?	Illustrative model: staff coordinate pickup/delivery by phone and confirm completion informally.
Exception handling	What happens if material is unavailable, dimensions change, rework is needed or delivery slips?	Illustrative model: the owner or job lead contacts the customer and updates the note/message; no actual practice is verified.

A3. Most frequent problem and impact

Evidence item	Current status
Most frequent operational issue	Not established. Illustrative hypothesis: difficulty seeing each job's current stage and next action.
Recent real example	No real incident was collected. Modeled example: a customer asks for an update while the job lead is away and the latest status is in a message or paper note.
Time cost	Not measured. A real audit should time the lookup and record the owner's estimate; no numeric saving is claimed.
Financial cost	Not quantified. No actual loss is attributed to the named business.
Current material/quantity control	Unknown. Illustrative control to investigate: material issued to a job, usable offcuts, scrap, rework and returned material should be recorded separately.
Direct quote 1	No actual quote collected. Illustrative dialogue only: “Customer ko status batane ke liye pehle kaam dekhna padta hai.” This is not a verified owner statement.
Direct quote 2	No actual quote collected. Illustrative dialogue only: “Agar agla step aur zimmedar aadmi dikh jaye, follow-up aasaan hoga.” This is not a verified owner statement.

Working hypothesis — not a finding For a job-based fabrication shop, possible areas to explore include job-status visibility, recording of measurements/changes, material leftovers/scrap, finishing/installation coordination and communicating delays. Ask open questions and report only what the owner confirms.

A4. Process-map photo and tracking evidence

After the interview, hand-draw the actual process with each stage, responsible role, hand-off and confirmed exception path. Add your own photos only with permission. Blur customer names, phone numbers, prices, addresses and other confidential information.

PROCESS MAP — NOT A FIELD PHOTO
Customer enquiry
↓
Measure job + confirm specifications
↓
Quote / approval / advance (if applicable)
↓
Create job card + promised date
↓
Material issue → cutting / welding / fabrication
↓
Finishing / outsourced work (if needed)
↓

Quality check + rework decision
↓
Delivery / installation → customer confirmation
↓
Invoice / payment follow-up

Exception loop: material shortage, changed specification, rework or delayed finishing → record blocker, owner and next action → resume job.

This is a modeled workflow for discussion. A real submission should include a hand-drawn map made after the actual interview.

SAMPLE TRACKING LAYOUT 1 — NOT A PHOTO OF THE BUSINESS
SAMPLE JOB REGISTER
Job ID: JOB-026
Customer: Sample Customer A
Job: MS gate frame
Specification: Dimensions and finish confirmed on job note
Stage: Fabrication
Promised date: 22 Oct (sample date only)
Blocker: None recorded in this mock-up
Next action: Complete welding and quality check
Owner: Fabrication lead
Last updated: 16 Oct (sample entry)

This mock-up demonstrates useful fields only; it is not a verified business record.

SAMPLE TRACKING LAYOUT 2 — NOT A PHOTO OF THE BUSINESS
SAMPLE JOB CARD / MATERIAL MOVEMENT
Job ID: JOB-026
Material issued: MS square tube — quantity and unit recorded at issue
Material returned: Usable offcuts recorded separately
Scrap/rework: Separate quantity and reason
Work stage: Welding → grinding → finish
Quality check: Pending / passed / rework required
Customer update: Date, channel and staff member recorded
Sign-off: Job lead and checker

No real material quantities are shown because none were observed. A real audit should photograph an actual tracking method only with permission and after blurring private information.

A5. Preliminary conclusion structure

Complete after the interview: “The business's main operational challenge is [verified issue]. Its current process uses [actual tools] and involves [actual roles]. The clearest evidence is [real example and/or direct quote]. The impact is [measured/owner-estimated amount, or ‘not quantified’]. A potential improvement is [small, practical change] because [evidence-based rationale].”

PART B — Engineer-Ready Requirements: Vardhan Steelware

Case context from the assignment: Vardhan Steelware is a stainless-steel utensils manufacturer in Jagadhri, Haryana, with about 70 staff and 40 distributors. Suresh Vardhan is owner, Kunal maintains the order sheet, Ramesh manages dispatch and a diary, and Munshi ji uses Tally.

B1. Problem statement and prioritization

Suresh's stated need is to check an order's status on his phone rather than call Ramesh to search a diary. Order information appears across a spreadsheet, diary and WhatsApp. Kunal requests a modern system, marketplace integrations and an AI sales dashboard, but online sales are stated to be under 5% of the business. Phase 1 should prioritize reliable order status, blockers, partial quantities and next-action ownership; advanced analytics and integrations should wait until the workflow and data quality are validated.

Case evidence	Requirement implication
About 60–70 orders/month; owner says the issue is knowing where goods are stuck.	Prioritize status and blocker visibility, not an assumed capacity problem.
Previous ERP cost ₹1.2 lakh in 2023; used about three months because workers found it complicated.	Adoption is the leading risk. Co-design a simple workflow with dispatch/shop-floor users.
Online sales are stated to be under 5%; Kunal reports 30% monthly growth.	Clarify the growth baseline and period before prioritizing integrations or forecasting.
Sharma order held for ₹1.8 lakh old dues despite a general statement that payments are not a problem.	Represent holds with reason, owner, date and release condition.

B2. Record-level issues and requirements

Supplied case evidence	Design response / validation
Gupta Kadai No.10, 200 pcs at ₹185, appears dispatched and pending.	Use a stable order ID and status history; flag conflict for human review.
Jain Kadhai No.10, 150 pcs at ₹192, appears twice as pending with a repeat-message note.	Capture source/timestamp; flag possible duplicates without auto-merging.
Mehra ordered 300 Kadai; 180 dispatched and remaining 120 later marked dispatched.	Track ordered, dispatched and remaining quantities at line level.
Sharma Gilas 30 dozen at ₹260 held for ₹1.8 lakh; separate Sharma Patila entry also on hold.	Represent hold reason, authorizer, owner and release condition.
Dispatched Thali line for 20 dozen at ₹600 has no customer; Verma Kadhai line is 180 kg with rate unconfirmed.	Controlled missing-customer exception; support rate-pending status.
Sharma Gilas return: −6 dozen due to dents.	Link return to original order/dispatch; capture reason and disposition.
420 kg sent to polishing contractor; 1,180 pieces returned; another message says 6 kg short.	Track batch and units separately. Do not compare kg with pieces without validated yield/conversion rules.

Step	Current picture from case	Visibility gap
Order entry	WhatsApp and sheet; Kunal maintains sheet.	Stable IDs, source and duplicate review.
Production / polishing	Some polishing outsourced to 4–5 contractors; material weighed out and goods counted on return.	Batch-to-order link, units and yield rules.
Dispatch	Ramesh keeps diary; sheet shows pending, dispatched and partial states.	Remaining quantity and latest status.
Holds / accounts	Payment holds; Munshi ji uses Tally.	Authoritative source and hold-release rules.

B3. Reconstructed current workflow (validate with client)

B4. Phase 1 scope and exclusions

Include in Phase 1	Defer / exclude and why
Unique order/order-line IDs; customer, product/specification, quantity/unit, order date/source, rate status and requested date.	AI forecasting and advanced dashboard: data quality and online-sales baseline need validation.
Status timeline with editor/time; blocker reason, owner, next action and follow-up date.	Full Amazon/Flipkart integration: validate actual volume and process needs after core status flow works.
Partial dispatch and balance quantity; returns linked to order; outsourced batch with input/output units and variance note.	Full ERP replacement or automatic Tally integration: prior ERP adoption failed; scope and data ownership unknown.

Mobile owner view; roles and change history; import preview and duplicate warnings.	Automatic duplicate merge or material-loss calculations across kg/pieces: unsafe without validation.

PART B — Data model, roles and reports

B5. Initial data model

Entity	Key fields / relationship
Customer	customer_id, name, region/type; one customer has many orders. Restrict contact data by role.
Product	product_id, name/code, specification/size, base unit, active status.
Order	order_id, customer_id, date, channel, created_by, requested date, overall status; has many lines/events.
OrderLine	line_id, order_id, product_id, ordered_qty, unit, rate, rate_status, line_status.
StatusEvent	event_id, order/line, old/new status, timestamp, changed_by, note.
Dispatch / DispatchLine	dispatch_id/date/reference; line-level dispatched_qty, unit, return_qty and reason. Multiple dispatches per order allowed.
ProcessingBatch	batch_id, contractor, order/line, sent date, input qty/unit, expected return, returned qty/unit, variance note, checked_by.
Blocker	blocker_id, order/line, type, description, owner, raised_at, next action, follow-up date, resolved_at.
User / Role	user_id, display name, role, active status; Owner, Order Admin, Dispatch, Accounts.

B6. Roles and phone view

Role	Access / action
Owner — Suresh	Mobile read view; search order/customer; see stage, last update, pending quantity, blocker, owner and next action.
Order admin — Kunal / assigned staff	Create/update orders and lines; capture source/due date; review duplicate candidates.
Dispatch — Ramesh	Update stage, record partial dispatch/balance, manage blockers and follow-up.
Accounts — Munshi ji	Confirm payment holds/rate status and access required commercial fields; no assumed Tally sync.

Proposed order card: Order ID • Customer • Product/quantity • Current stage • Last updated by/at • Remaining quantity • Blocker/reason • Next action/owner • Expected dispatch date only if confirmed. Show 'not provided' rather than inventing an ETA.

B7. Three morning reports

Report	Fields	Action enabled
1. Blocked / on-hold orders	Order, customer, reason, owner, days open, next action, follow-up date.	Assign or escalate next action; route payment holds to authorized staff.
2. Due / overdue / partial dispatch	Requested date, stage, ordered/dispatched/remaining qty, latest update, dispatch owner.	Prioritize at-risk orders and clarify balance quantities.
3. Outsourced processing / reconciliation	Contractor, batch/order, sent date/unit, expected return, returned qty/unit, variance note, reconciliation state.	Follow up overdue returns and investigate differences without assuming unlike units are comparable.

B8. Discovery questions

Question	Stakeholder
What are the exact order stages, and who may change each status?	Suresh + Ramesh
How do we distinguish a repeated message from a new order or status update?	Kunal
Which fields are mandatory at order acceptance: customer, product, unit, rate, due date, payment terms?	Kunal + Munshi ji
How are partial dispatches, returns, replacements and cancellations linked to the original order?	Ramesh + Kunal
How are polishing input/output weight, pieces, scrap and rejected goods reconciled?	Ramesh + Munshi ji
What is the authoritative payment-hold source, and what should Phase 1 read from Tally?	Suresh + Munshi ji
What made the previous ERP difficult, and what devices/connectivity do users have?	Ramesh + workers
Which exceptions should appear first each morning, and who approves the pilot and target date?	Suresh + Kunal
What baseline supports the 30% online growth figure and over what period?	Kunal

B9. Biggest risk and mitigation

Risk — low adoption. The earlier ERP was abandoned after about three months because it was considered complicated. A technically capable system will fail if people do not update it.

- Co-design the status flow with Ramesh and representative shop-floor users.
- Keep data entry minimal and reuse order-line details rather than re-entering them.
- Prototype the phone view and dispatch update flow with real example orders.
- Pilot a small order set and compare statuses against the current sheet/diary.
- Provide short role-based training, define an owner for the process, and keep export/fallback available.

B10. Pilot acceptance checks

Check	Pass condition to agree with client
Status visibility	Owner can find an order on phone and see stage, last update, blocker and next action.
Partial dispatch	User records a partial dispatch and system shows balance quantity in the same unit.
Payment hold	Authorized user records hold reason/release condition; owner sees the hold without unnecessary sensitive detail.
Outsourced batch	Input/output quantities and units are stored separately with a variance note and follow-up owner.
Duplicate candidate	Possible duplicate is flagged for review; no silent merge or deletion.

Before committing Confirm status vocabulary, unique IDs, unit/yield rules, payment-hold authority, Tally source of truth, pilot users, acceptance criteria and the exact date meant by 'before Diwali'. Do not promise delivery before discovery.

PART C — AE-to-Engineering Handoff

Handoff area	Summary
Decision maker	Suresh Vardhan (owner; signs cheques). Wants to check order status on his phone.
Influencers / users	Kunal maintains the spreadsheet and requests integrations/AI dashboard; Ramesh manages dispatch and a diary; Munshi ji uses Tally.
Core pain / evidence	Status information is split across sheet, diary and WhatsApp. Case examples include conflicting Gupta status, repeated Jain row, partial Mehra dispatch, Sharma payment hold, missing customer and polishing reconciliation.
Request vs. first need	Requested: modern system, marketplace integrations, AI dashboard. First need: reliable status, blocker ownership, partial quantities and simple mobile visibility.
Risks / response	Previous ERP adoption failed due to complexity. Co-design a simple status flow, pilot with actual users and defer advanced integrations until data quality/adoption are proven.
Next step / timeline	Discovery call with all four stakeholders to validate statuses, units, holds, reports and pilot acceptance. Confirm exact 'before Diwali' date; no delivery promise before discovery.

Engineering request Validate the current workflow and data definitions with the named users before estimating effort. This handoff intentionally contains no price or unsupported delivery commitment.

PART D — Marketing Content

PART D BASIS AND LIMITATION
The public post and ad are complete drafts based on the modeled workflow scenario. They do not claim to report verified findings from SS MS Fabrication Works. Before publishing as a genuine audit insight, conduct the required interview, confirm a real process gap, remove identifying details, and obtain any required permission for images.

D1. LinkedIn post (maximum 150 words)

A modeled workflow suggests that job-status updates can be spread across calls, messages and paper notes. This is a discovery hypothesis, not a verified finding about the named workshop. The example illustrates why each job needs a current stage, blocker, responsible person and next action. No time or money impact was quantified because no field interview or timing observation was performed.

The lesson: before proposing a big system, understand how work actually moves. Map the steps, identify who owns each update, make exceptions visible, and design around the tools people will realistically use.

Good solutions engineering starts with listening and evidence—not assuming every business needs a complex platform.

#SolutionsEngineering #Operations #CustomerDiscovery

D2. Meta ad concept

Element	Draft / decision
Language	English — selected for a broad professional B2B audience; wording is plain and avoids unsupported promises.
Primary text	Still tracking customer jobs across calls, messages and paper notes? A simple job-status view can help a team record the current stage, blocker, next action and who owns the follow-up. Start by mapping the real workflow before choosing software. No savings or performance claims are made.
Headline	Make Every Job's Next Step Clear
Creative concept	A clean split visual: left side shows generic paper notes and message bubbles; right side shows a simple job list with columns for Job, Stage, Blocker, Owner and Next Action. Use fictional sample data only; do not use real business records or identifiable photos.
Accuracy / privacy	Keep the audited business anonymous. Do not name Vardhan Steelware or claim verified savings, guarantees, or features of a product that is not actually being promoted.

D3. How Part A should inform the final copy

- Choose one specific insight actually supported by the interview and observed process.
- Use a number only if the owner provided it or you measured it; label estimates clearly.
- Do not use the business name, identifiable photos, customer records or owner quote in public marketing content.
- If no financial impact was quantified, say so instead of inventing a savings claim.

Source notes and final checklist

Public listings used only to identify a possible audit subject (accessed 16 October 2026):

- SS MS Fabrication Works listing with address and service categories: https://in.infoaboutcompanies.com/Catalog/KA/Bengaluru/Metal-Fabricator/SS-MS-Fabrication-Works
- Related directory listing for SS MS FABRICATOR'S Works with address and public contact: https://in.infoaboutcompanies.com/Catalog/KA/Bengaluru/Welding-Supply-Shop/SS-MS-Fabrication-Works (confirm current details before relying on it).
- Alternative directory entry and local workshop context: https://www.justdial.com/Bangalore/Metal-Fabricators-in-Rk-Puram/nct-10319476

Directory details may be stale or inconsistent. Confirm the trading name, address, activity and contact directly with the business before the visit. These public sources do not verify interview-only facts.

DOCUMENT STATUS

Parts B–D are drafted from the supplied Vardhan Steelware case and the modeled workshop workflow. The KR Puram business address comes from a public directory listing. A real owner interview, direct quotations, on-site observations and permission-based photographs were not supplied, so those items are not represented as verified facts. Complete those specific field-evidence requirements before submitting this as a real audit.